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Education

New York University, Abu Dhabi

Sep. 2021 – May 2025

Bachelor of Science in Computer Science & Minor in Mathematics

Abu Dhabi, UAE

Cumulative GPA: 3.94/4.00

Honors & Awards: Magna Cum Laude, NYU Abu Dhabi Founder's Award (Highest Honors)

Publications:

- Neutralizing Narratives: Debiasing News Articles with AI (First Author, Under Review at ICWSM'26)
- Does 4-bit Weight-Only Post-Training Quantization Preserve BOS-Sink Head Topology? An Empirical Study for Edge-Deployed Small Language Models (Co-author, Under Review at IEEE IECON 2026)
- ARID: A Deployable Edge AI System for Structured Information Extraction from Industrial Maintenance Work Orders (Co-author, Under Review at IEEE IECON 2026)

Work Experience

GitRoll Technology Limited

Oct 2025 – Now

Co-founder & Founding Engineer

Remote

- Architected and led development of GitRoll's SkillsGraph Analyzer, the flagship engine behind its "proof-of-skill from real code" thesis: maps a developer's actual git commits to skill checkpoints across 62 technical roadmaps via a multi-stage pipeline (repository scan, LLM roadmap filtering, Vertex AI embedding + semantic search + Gemini contributor analysis), with a Karma Points pipeline that scores meaningful contribution through git-blame attribution, a 3-level noise filter isolating the most impactful 5% of lines, and multi-factor weighting across file type, difficulty, code age, and language.
- Shipped the SkillsGraph Analyzer end-to-end across all 4 monorepo layers — Postgres/Prisma schema, server-side scan integration, a contributor-matrix API, and Next.js dashboards with interactive contributor and cumulative-impact visualizations.
- Owned and directed the Assessment Engine, GitRoll's proprietary AI auto-grading API for a federal Treasury/IRS workforce contract: led a 4-engineer team building 3 parallel grading pipelines (deterministic rubric, SkillsGraph capability extraction, AI-session-log analysis) toward a federal ship gate.
- Single-handedly built a custom Claude Code automation layer for company operations — 14 skills, 9 agents, and 4 enforcement hooks — including a cross-platform sync engine and an unattended meeting-processing pipeline that processed 300+ transcripts across a self-organizing 30,000+ file knowledge base.
- Drove client-facing BD and technical discovery for 10+ enterprise prospects across Taiwan financial-holding companies, US workforce platforms, and APAC healthcare/biopharma — sourcing leads, running demos, and scoping engagements — and structured the company's first paid client retainer.

PicCollage

June 2025 – Nov 2025

AI Engineer

Taipei City, Taiwan

- Redesigned PicCollage's HRIS system with 4 interconnected databases to securely manage 300+ employees' data, automating cross-platform synchronization and reducing HR administrative overhead by 70%.
- Architected automation workflows that integrated 4 platforms, including Amplitude and AppFigures, to eliminate manual data syncing and saving 50 minutes per iteration.
- Co-developed AI agents for dynamic bid optimization on Apple Search Ads, automatically adjusting keyword bids based on real-time CPA/ROAS metrics and improving ad spend efficiency across 200+ campaigns.

Taiwan Semiconductor Manufacturing Company (TSMC)

June 2024 – Sep 2024

Machine Learning Intern

Tainan, Taiwan

- Implemented an XGBoost model that enhanced TSMC's production forecast accuracy by 4%, replacing TSMC's traditional prediction model and directly impacting over 10,000 products and 150+ customers.
- Developed a robotic process automation (RPA) pipeline with SQL, Excel, and PowerBI to automate the data extraction, transformation, and loading (ETL) process, reducing data processing time by 30%.

Metamory Technology Limited

May 2023 – Jan 2024

Web Development Intern

Hong Kong, China

- Built and instrumented a React.js marketing site with 13 GTM tags integrated into GA4 for full-funnel user analytics, and ran targeted Google Ads campaigns that drove a 160% surge in web traffic.

Research

NYU Tandon’s Flexible AI-enabled Mechatronics Systems Lab (FAMS)

June 2024 – May 2025

Head Researcher & Product Manager

New York Hybrid, USA

- Led a team of 9 to build an AI-driven educational platform connecting students and faculty to 11 specialized experts across academic and wellness domains.
- Designed a multi-layer semantic-routing framework that directs each query to the right expert in $O(\log n)$, reaching 95% routing accuracy under 50ms.
- Built RAG retrieval over a ChromaDB vector store of 184 documents (5,000+ pages of course materials and wellness resources), mitigating hallucinations via vector embeddings.
- Deployed FastAPI microservices on Kubernetes within NYU’s HSRN with SSO authentication, scaling to 10,000+ user messages.

NYU Abu Dhabi’s Data Science and AI Lab

Feb 2024 – May 2025

Lead Researcher

Abu Dhabi, United Arab Emirates

- Studied the racial and gender bias trends in 10 popular news publishers with over 355k articles from the past decade.
- Leveraged 6 state-of-the-art LLMs for bias detection and removal, employing few-shot learning and ensemble methods.
- Designed a large-scale crowdsourced study using randomized controlled trials to assess the impact of debiased news on readers’ perceptions, applying causal inference techniques for analysis.

Technical Skills

Languages: Python, TypeScript, JavaScript, C/C++, SQL
ML/AI Stack: PyTorch, TensorFlow, LangChain, RAG, Agentic AI, ChromaDB, Harness Engineering
Backend & Web: FastAPI, Node.js, Express, Next.js, React, PostgreSQL, Prisma, MongoDB
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Git, n8n
Data: Pandas, NumPy, Excel, PowerBI

Certificates & Contests

TSMC Regional Intern Competition Gold Award (4/250 Interns) TSMC

August 2024

IBM Data Science Professional Certificate IBM

January 2024

Deep Learning Professional Certificate DeepLearning.AI

July 2023

American Region Mathematics League International 3rd

June 2020

Projects

Savory Genius - AI Meal Planner | React, TypeScript, Express, PostgreSQL, OpenAI, Vercel

2024

- Built a full-stack web application using React 18, Express.js, and PostgreSQL with Drizzle ORM for AI-powered meal planning, recipe management, and nutritional analysis.
- Integrated OpenAI GPT-4o-mini to generate personalized recipes based on available ingredients, analyze meal history for nutritional insights, and provide dietary recommendations.
- Deployed serverless architecture on Vercel with automated shopping list generation, calendar-based meal scheduling, and multi-tenant user isolation using Clerk authentication.

LinkedIn Weak Ties AI Assistant | Python, FastAPI, React, ChromaDB, Azure OpenAI, Tailwind CSS

May 2025

- Built a two-tier semantic search system that identifies top 4 relevant connections from 1000+ LinkedIn profiles in under 100ms using multi-attribute vector search and GPT-4.1 refinement.
- Developed full-stack application with ChromaDB vector store using weighted similarity scoring across 4 profile dimensions (summary, position, location, industry) for intelligent connection matching.
- Integrated AI-powered outreach generator that creates personalized LinkedIn reconnection messages based on enriched profile context and user business goals.

Online Telehealth Portal - Smartel | Amazon AWS, FastAPI framework, React, PyDantic, Typescript

Jan 2024

- Built Smartel, a telehealth portal that lets users schedule appointments; record, transcribe, and summarize consultations; and deliver concise session reports to physicians.
- Led the backend development, implementing robust FastAPI and integrating OpenAI technologies (Whisper and GPT-4) to enable seamless conversation transcription and automated consultation summary generation.
- Designed a scalable database with Amazon AWS and backend architecture with PyDantic, FastAPI, SQLAlchemy, and PostgreSQL. Capable of handling 1000+ concurrent users, 500 bookings per minute, and response time of under 3 seconds for data retrieval operations.